



**University of
East London**

**SCHOOL OF ARCHITECTURE, COMPUTING
AND ENGINEERING**

**Engineering
Postgraduate Courses**

EG7030

MSc Dissertation

**Module Leader
(Dissertation Coordinator)
Dr Ali Abbas**

**STAFF RESEARCH AREAS / SAMPLE DISSERTATION
TOPICS**

SELECTION OF DISSERTATION SUPERVISOR AND TOPIC

Students **must read the Dissertation Handbook** (which can be found online on Moodle or Moodle for the module) for full details on the dissertation requirements, format, submission dates, Turn-it-in and other regulations.

Before progressing to the dissertation, students registered on the MSc are expected to have **passed the majority of the taught modules** particularly in the subject area of the proposed dissertation. Students **must also attend the mandatory Research Method classes** (failure to do so will result in submission not been accepted).

To qualify for the award of the MSc, students must complete a satisfactory dissertation of approximately 12,500 words. The objective of the dissertation is to develop the students' ability to study **independently**, making their **own critical appraisal** of the chosen subject and drawing from this the appropriate conclusions.

Deadlines for choosing a topic and an appropriate supervisor are published in advance to give the students sufficient time to contact their potential supervisor and agree a topic. Please refer to the **Important Dates List** provided at the beginning of this handbook. Students **must register by the specified deadline** quoting the name of their supervisor on the registration form. Any registration form without a supervisor name **will not be processed**. For the **latest list of supervisors**, their research interests and contact details, the staff section on the School website should be consulted. A list of topics and research areas and groups is also provided below.

Students who wish to complete their dissertations over semester C (summer period) are advised to contact their supervisors and agree their supervision **well before** the deadline. Semester C is available for research only in the case that it has been agreed by the research supervisor.

If you are having difficulties in finding a supervisor, please let the module leader know **as early as possible**.

Students are expected to spend **at least 600 hours** (60 C) on the dissertation. Further information on the requirements of the dissertation module are provided in the **module handbook and specifications** and the students are strongly advised to consult these two documents as early as possible.

One or more members of the civil engineering and surveying field will supervise the dissertation work of each student. Students **must** undertake their research investigations in the **Docklands Campus**, unless work with an outside employer or authority is involved. The involvement of an outside employer or authority will require the agreement of the students' departmental supervisor and the Module Leader, and will require an external supervisor who will liaise with the departmental supervisor to confirm the authenticity of the student's dissertation work.

Prior to commencement of the dissertation, each student must submit a project **proposal** with a **time management plan**. This will be developed in conjunction with the project supervisor, who will judge the dissertation's worthiness as MSc standard. This proposal sets out the work that will be carried out for the dissertation. The research proposal must also set out the problem and the methodology to be adopted and be a feasible project to carry out in the time available.

Students who wish to defer their dissertations must seek the approval of the Programme Leader and ask for an intermission, and inform the Module Leader.

Research Topics and Supervisors

Students will be required to demonstrate that the research undertaken has been completed to an appropriate level for a Masters award. The dissertation must therefore, in general terms, include elements of research, independent work, the derivation of a product (through laboratory testing, critical analysis or computer programming) and the analysis of data.

The research needs to make a contribution to the understanding of the subject area studied.

Sample research areas of the academics involved in supervision of MSc dissertations are shown in the following table **simply to indicate** their expertise and interest (but please note that the **samples shown might not necessarily be available, so you need to check with relevant staff regarding their latest topics**):

*Please contact supervisors from your **Area** below only (based on your MSc course).*

Area	Academic	Sample research areas
Structures & Construction Materials	Dr Ali Abbas abbas@uel.ac.uk	• Finite-element analysis (concrete structures) • Experimental studies (low-carbon concrete, bio-concrete) • Ground floor slabs • Fibre-reinforced concrete structures
	Dr Jawed Qureshi j.qureshi@uel.ac.uk	• Steel-concrete composite structures • Fibre reinforced polymers (FRP) • Steel structures • Finite-element analysis
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Geotechnical Engineering	Dr John Walsh j.w.walsh@uel.ac.uk	• Piled foundations including pile groups and lateral load response • Effect of Eurocode 7 on geotechnical design • Use of design software packages in geotechnical design • Design of reinforced earth structures

	Dr Arya Langroudi A.AssadiLangroudi@uel.ac.uk	• Experimental studies. • Urban Geotechnics • Slope stability • Tunnelling Engineering
Hydraulics /Coastal Engineering		•
Transportation & Highway/Railway Engineering / ITS		•
	Dr Rasha Mohammad Rasha.Mohammad@uel.ac.uk	• Transportation Logistics • Railway Engineering • Ports
	Dr Khalid Abdalla kmaa4@cantab.ac.uk	• Transportation Engineering / ITS. • Geotechnical Engineering.
	Sami Khairi S.Khairi@uel.ac.uk sami.khairi@live.co.uk	• Transportation Engineering. • ITS/ AI applicaions.
Mechanical Engineering		•
	Dr Elnaz Naghibi E.Naghibi@uel.ac.uk	• Mechanical Engineering • Engineering modelling
	Dr Mohammad Rabbani M.Rabbani1@uel.ac.uk	• Mechanical Engineering • Biomedical Engineering • Engineering Design • 3D Modelling • Total hip joint replacement
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	Dr Abdulrahman Albar A.Albar@uel.ac.uk	• Mechanical Engineering / Product design. • Electric vehicle technologies. • AI and automation/ Robotics control • Renewable energy systems
	Juan Bilbao-Ludena J.Bilbao-Ludena@uel.ac.uk bilbaosjc@gmail.com	• Mechanical Engineering • Aerospace Engineering
	Dr Muhammad Anwar M.S.Anwar@uel.ac.uk	• Mechanical Engineering • Mechatronics • Sustainable Energy /Renewable Energy Systems
Engineering /Mechatronics/Electro-Mechanical/Energy/Aerospace		•
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	Dr Komal Saleem K.Saleem@uel.ac.uk	• Renewable Energy Systems • Sustainable Energy Generation Technologies • Electric vehicles (V2G/X)
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	Dr Muhammad Anwar M.S.Anwar@uel.ac.uk	• Mechanical Engineering • Mechatronics • Sustainable Energy /Renewable Energy Systems
	Mayank Trivedi M.Trivedi@uel.ac.uk	• Electrical Engineering. • Cloud Engineering, Technical Engineering, Software Development, Data Engineering.
	Dr Abdulrahman Albar A.Albar@uel.ac.uk	• Mechanical Engineering / Product design. • Electric vehicle technologies. • AI and automation/ Robotics control • Renewable energy systems
	Juan Bilbao-Ludena J.Bilbao-Ludena@uel.ac.uk bilbaosjc@gmail.com	• Mechanical Engineering • Aerospace Engineering

	Mudita Juneja M.Juneja@uel.ac.uk	Electrical Engineering
Engineering Management		•
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	Thilakshi Hiyare Gamage t.hiyare-gamage@uel.ac.uk	Engineering Management
	Gaurav Kirti Patel G.Patel1@uel.ac.uk	Engineering Management
	Nilakshan Balasubramaniam n.balasubramaniam@uel.ac.uk	Engineering Management
	Kaveesh Abeysuriya K.Abeysuriya@uel.ac.uk	Engineering Management
	Vinod Bashyal V.Bashyal@uel.ac.uk	Engineering Management
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	Dr Hafiz Farahat H.Farahat@uel.ac.uk	Engineering Management
	Buddhika Premaratne B.Premaratne@uel.ac.uk	Engineering Management
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	Durfashan Tariq er.durfashan002@gmail.com	Engineering Management
	Anupam Dey anu3dey@gmail.com	Engineering Management